Morphological Transformation and Spatial-Logical Aggregation for Tree Species Classification Using Hyperspectral Imagery

Mengmeng Zhang, Wei Li, *Senior Member, IEEE*, Xudong Zhao, *Student Member, IEEE*, Huan Liu, Ran Tao, *Senior Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Hyperspectral image (HSI) consists of abundant spectral and spatial characteristics, which contribute to a more accurate identification of materials and land covers. However, most existing methods of hyperspectral image analysis primarily focus on spectral knowledge or coarse-grained spatial information while neglecting the fine-grained morphological structures. In the classification task of complex objects, spatial morphological differences can help to search for the boundary of fine-grained classes, e.g., forestry tree species. Focusing on subtle traits extraction, a spatial-logical aggregation network (SLA-NET) is proposed with morphological transformation for tree species classification. The morphological operators are effectively embedded with the trainable structuring elements, which contributes to distinctive morphological representations. We evaluate the classification performance of the proposed method on two tree species datasets, and the results demonstrate that the proposed SLA-NET significantly outperforms the other state-of-the-art classifiers.**

***Index Terms*— Tree species, convolution neural network, deep learning, hyperspectral image (HSI), morphological transformation.**

I. Introduction

IN HYPERSPECTRAL image (HSI), a pixel records the electromagnetic reflectance over a wide spectral range, which enables diagnostic cognition of different materials at a fine-grained level [1], [2], [3], [4]. Also, HSI provides a new capacity for acquiring land cover information on the Earth's surface [5], [6], [7], [8]. For example, fine classification of the fragmented tree species distribution has become an important application in recent years, and it has been demonstrated to be crucial for forest resource inventory, forestland carbon storage estimation, and species diversity monitoring [9].

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Mengmeng Zhang, Wei Li, Xudong Zhao, Huan Liu, and Ran Tao are with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: mengmengzhang@bit.edu.cn; liwei089@ieee.org).

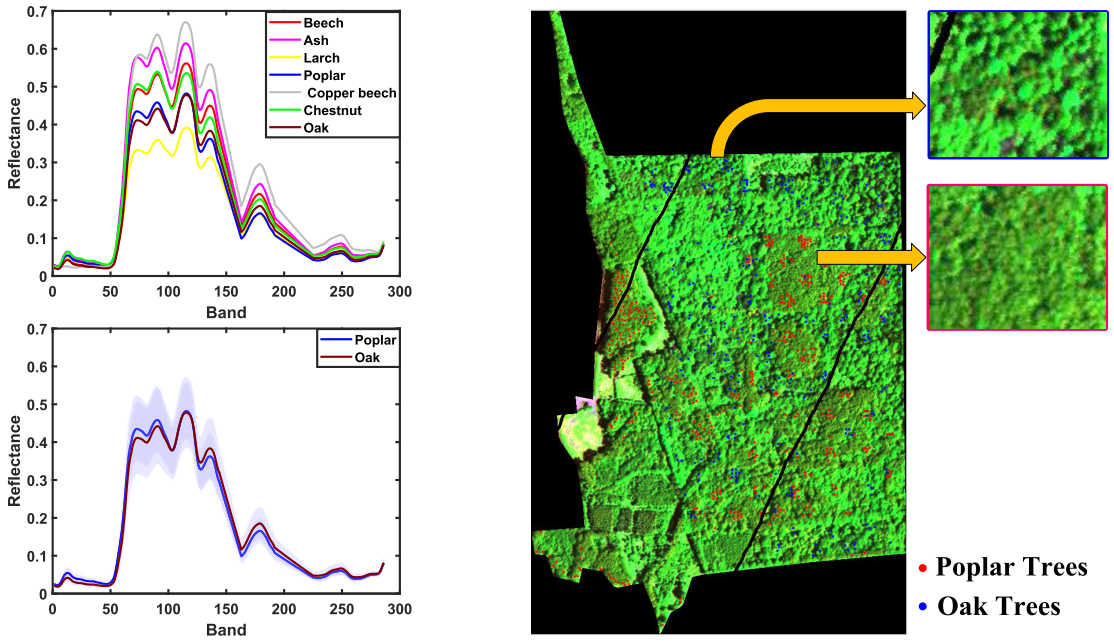
Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Mississippi State, MS 39762 USA (e-mail: du@ece.msstate.edu).

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Supervised classifiers are commonly used in the pixel-level classification task. For example, support vector machine (SVM) [10], extreme learning machine (ELM) [11], and random forest (RF) [12] was investigated in some research. Particularly, RF is a widely used machine learning algorithm for tree species classification, and it is designed based on simple decision tree models. Obviously, it is possible to discriminate classes with spectral information; however, spectral uncertainty [13], [14] is still an intractable issue in the HSI processing task. It is necessary to integrate spatial information to supplement the knowledge provided by HSI instruments [15], [16], [17], [18]. Although many spectral–spatial techniques have been investigated [19], [20], [21], these methods rely heavily on prior conditions and expert knowledge, leading to poor portability and unreliable performance [22], [23]. On the other hand, deep learning methods [24], [25], [26], [27], [28], which realize the goal of dynamic feature extraction, have been widely utilized in remote sensing image classification tasks. For example, Mei et al. [26] extracted spatial–spectral features based on the well-designed convolutional neural network (CNN), and these features can better utilize the local contextual information for HSI classification. Lee and Kwon [27] proposed a contextual deep CNN (CD-CNN) for spatial–spectral information aggregation, and multiscale convolutional filter banks were specially designed to optimize the spatial-textures' utilization. Different from conventional deep networks that utilize a spatial convolution kernel to extract features [26], [27], [29], Zhang et al. [28] further developed a diverse region-based CNN for discriminative-appearance factors' acquisition, which has yielded significantly better results than other techniques. Beyond these CNN-based schemes, recurrent neural network (RNN) [30], generative adversarial network (GAN) [31], [32], [33], capsule network (CapsNet), and graph learning [34], [35] were gradually utilized in HSI classification. Taking the CapsNet as an example, a novel spatial–spectral capsule network was proposed in [36], and this method was demonstrated to be effective in improving classification accuracy. Also, based on CNN and graph convolutional network (GCN), Hong et al. [34] gave a comprehensive comparison between GCN and CNN on the HSI classification task. Moreover, the attention mechanism has been introduced in HSI analysis to improve the category discrimination accuracy effectively [37],

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(a) (b)

Fig. 1. Representation of spectral reflectance and spatial contextures of different classes in Belgium data. (a) Mean spectral reflectance of different tree species. (b) Spatial contextures of different tree species (poplar trees are partly marked with red spots, and oak trees are marked with blue spots).

[38]. Zheng et al. [38] proposed a rotation-invariant attention network (RIAN) to effectively deal with the spatial rotation of HSI, which has shown excellent and superior performance in HSI classification.

Although the aforementioned methods performed well in the rough-class classification of HSI, they are less effective when the spectral responses of different classes are similar while spatial textures of the same class vary considerably. As shown in Fig. 1, the mean spectral curves of different tree species are presented, and it is clear that the spectral waves represent an aliasing state. Particularly, the mean and standard deviation of spectral curves for oak trees and poplar trees are depicted in the bottom of Fig. 1(a), and the serious multipeak overlap phenomenon further reveals the difficulty of tree species classification from the spectral perspective. While in Fig. 1(b), there are micromorphological gaps between poplar trees and oak trees: the contextures and morphological structures for poplar trees are obviously more even and compact than that for oak trees. In fact, the spatial distributions displayed in satellite images are consistent with the biological characteristics of tree species. Stong, narrow, and tall are usually the typical characteristics of poplar trees, which contribute to the even distribution patterns of the poplar-species group. On the other side, oak trees grow to different heights within a wide range, while their crowns (all the branches and leaves) tend to grow to an impressive size. With the aforementioned features, the oak-species group usually takes on a broad round-like look, and it inevitably shows uneven textures from the satellite view. Therefore, the morphological differences among different tree species have the potential for enhancing the inter-class discrimination for tree species classification. In recent years, the approach of feature extraction realized by mathematical morphology has drawn great attention, and there are a considerable number of contributions in the literature on the use of morphological operations. For example, Guo et al. [39] assessed the ability of a morphological feature-based tree species identification approach to map 12 subtropical woody perennial species in the Hong Kong urban area. The axisymmetric and nonaxisymmetric canopy-based morphological features have shown satisfactory identification results in this work. Unlike in [39], another possible set of morphological approaches is constructed by applying a set of reconstruction operations with the structuring elements. For example, in [40], local contain profiles (LCPs) were firstly extracted based on the topology tree, and a dual-concentrated network was then applied to classify the LCP features. Based on these examples, it can be seen that existing morphological-based methods have proven to be effective in extracting informative fine-grained features from images. However, they still have limitations. For example, the parameters of structuring elements are fixed, and the feature extraction process is subject to the initial choice of the structuring elements, which can downgrade the crucial morphological characteristics acquisition and affect subsequent classifier performance.

In summary, the fine-grained classification of tree species is still a challenging task, not only because the spectral response of different plants in the same family and genus are extremely similar [22], but also because the differences in spatial contextures of various plants are relatively slight, and these crucial morphological characteristics are easily broken in the complicated feature extraction process. In this regard, the limitations of existing methods are summarized as follows.

1) For traditional methods, human-engineered parameter adjustment is tedious, and the characterization ability of spectral nonlinearity is limited.

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2) For the deep learning-based models, there are inevitably misalignments between the network's focus and the real objects in an image, which leads to a critical structural loss in the information propagation process. To be specific, the detailed information related to object boundaries is missing due to the regular downsampling or convolutions with striding or pooling operations [37], which will lead to distortion in the classification map, therefore decreasing final classification accuracy.

Focusing on the limitations of existing research, the classification strategy of tree species is improved from the perspective of diagnosis spectrum utilization and morphological characteristics retention. This article proposes a spatial-logical aggregation (SLA) network with the morphological transformation design to reduce the crucial structure loss in forestry information extraction and classification task. The main contributions of this article are summarized as follows.

1) An end-to-end multitunnel aggregation network is designed with a focus on the fine-grained tree species classification, which assembles spectral, spatial, and morphological details based on several plug-and-play dynamic-fusion modules.
2) Convolution is a linear operation, which is difficult to capture nonlinear characteristics like shapes and sizes of objects [41]. In this regard, trainable transformations are carefully designed to learn appropriate structuring elements based on the combination of both nonlinear morphological operations and linear convolutions.
3) To further preserve the desired fine spatial details, the multistream morphological feature extraction module is deployed with both dual-mode transmission and self-attention guided feature fusion. In particular, the dual-mode setting contains two complementary tunnels of morphological layers to perform the morphological operations. The combination is beneficial to improve the spatial perception ability of the whole model, which assists in more accurate classification of forestry species.

II. Proposed Classification Framework

The proposed classification pipeline is shown in Fig. 2, which consists of several key components, i.e., different feature extractors and an adaptive-feature-fusion module. Specifically, the proposed SLA-NET mainly guides the spatial-related extractors (spatial extractor and morphological extractor) to collect the spatial-logical features of tree species. Based on this design, features with different physical and semantic information are integrated by the following fusion module, and diversified context features are collaboratively fed into the final classifier.

A. Features Extraction

The classification framework starts with two kinds of feature extractors, and the extracted features with spatial and morphological information are denoted as $\mathbf{F}_{\text{spa}}$ and $\mathbf{F}_{\text{mor}}$, respectively. The individual design of different extractors is provided below.

1) Spatial Extractor: Due to the fact that the same land-cover class contains varied spectral signatures while different classes may share similar spectral responses, it is difficult to accurately differentiate different classes with only spectral knowledge [13]. Accordingly, to utilize spatial information and a minimalist set of encoded spectral information in HSI, the spatial extractor $\mathbf{E}_{\text{spa}}$ is designed to collect coarse-contextual information of HSI input. The spatial extractor consists of two parallel tunnels. As shown in part I of Fig. 2, the detailed settings of the 2-D processing tunnels are illustrated. In each processing tunnel, the continuous operation of Conv-BN-LeakyRelu-Conv-LeakyRelu-MaxPooling [42], [43] is executed, and corresponding features derived from the aforementioned operations are fused in succession. Also, in the implementation process, all the convolution operations are executed with zero padding, and the convolution stride is set as 1. Besides, the number of convolution kernels used in 3×3 convolution and 1×1 convolution operations is 64 and 128, respectively. To be specific, given a spatial patch $\mathbf{X} \in \mathbb{R}^{m \times n \times d}$, the feature embeddings obtained by the spatial extractor are formulated by

$$\mathbf{F}_{\text{spa}} = \mathbf{E}_{\text{spa}}(\mathbf{X}). \tag{1}$$

2) Morphological Extractor: For the fine-grained classification task of tree species, the spectral response of different plants is extremely similar [22], while the differences in spatial contextures of varied plants are relatively slight. Meanwhile, some key morphological characteristics are easily broken in the regular feature extraction process. In this regard, the core-structure learning of tree species should be reconsidered and redesigned from the perspective of morphological details retention. Compared with the full convolution framework, the proposed morphological extractor $\mathbf{E}_{\text{mor}}$ consists of both convolution operations and morphological transformations (erosion and dilation). Specifically, the integrated extractor is called the morphological transformation (M-T) module, and the detailed settings of this module are presented in Fig. 3. The M-T module contains several key components: 1) multistream morphological information exploration for extracting plentiful morphological structures; 2) polymorphic transformer to transmit dual-mode representations; and 3) dynamic fusion of features derived from different streams.

Once the input $\mathbf{X}$ passes through the M-T module, the low-level features $\mathbf{X}'$ are first extracted by two convolution layers. The number of these two convolution kernels is 32 and D, respectively. Here D denotes the channel number of the input HSI data. Then, three parallel streams are carefully designed to consolidate the high-resolution features with both convolutional and morphological operators. In each stream, the polymorphic transformer is equipped to extract features, and detailed settings are also illustrated in Fig. 3. The morphological operators are generally useful for mining the associations on the shape, size, and connectivity of different objects inside the image. Mondal et al. [44] pointed out that these associations are generally excavated by selecting structuring elements with suitable shapes and sizes. Accordingly, the optimized structuring elements play a crucial role in acquiring features that can better characterize the geometrical correlations of

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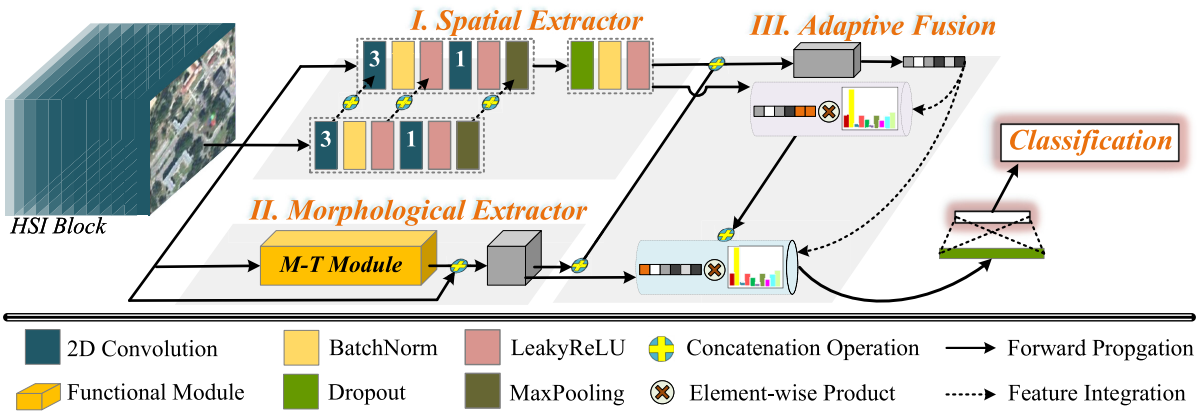


Fig. 2. Illustration of the proposed classification pipeline, including two kinds of feature extractors, adaptive fusion module, and classification process.

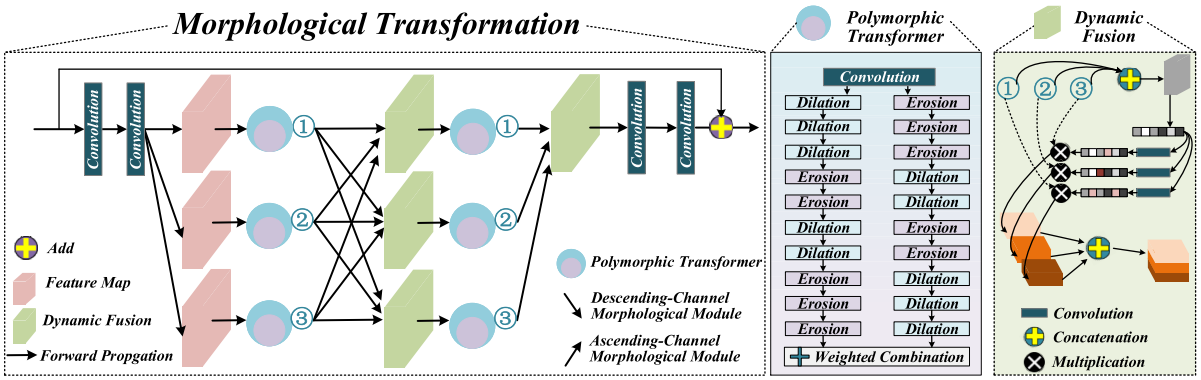


Fig. 3. Architecture illustration of the morphological transformation (M-T) module, including polymorphic transformer, descending-channel morphological module, ascending-channel morphological module, and dynamic fusion module.

objects inside the image, and these structuring elements can be learned from the backpropagation process. Following these principles, the polymorphic transformer is constructed using the dilation and erosion operating units. Specifically, for the input $\mathbf{X}'$, the dilation ($\oplus$) and erosion ($\ominus$) operations are calculated as

$$\begin{cases} \left(\mathbf{X}' \oplus \mathbf{W}_d\right)(x,y) = \max\limits_{\substack{i\in\mathbf{S}_1 \\ j\in\mathbf{S}_2}} \left(\mathbf{X}'(x-i,y-j) + \mathbf{W}_d(i,j)\right) \\ \left(\mathbf{X}' \ominus \mathbf{W}_e\right)(x,y) = \min\limits_{\substack{i\in\mathbf{S}_1 \\ j\in\mathbf{S}_2}} \left(\mathbf{X}'(x+i,y+j) - \mathbf{W}_e(i,j)\right) \end{cases} \tag{2}$$

where $\mathbf{W}_d \in R^{a\times b}$, $\mathbf{W}_e \in R^{a\times b}$, $\mathbf{S}_1 = \{1,2,\ldots,a\}$, and $\mathbf{S}_2 = \{1,2,\ldots,b\}$. $\mathbf{W}_d$ and $\mathbf{W}_e$ are the structuring elements, and related kernels implement corresponding operations in a window of size $a \times b$. All the structuring elements are initialized randomly in the convolutional kernel. For the polymorphic transformer, two complementary tunnels of morphological layers are adopted here. Further, following the weighted combination strategy proposed in [45], the integrated representations are generated as

$$\mathbf{X}_{\text{poly}} = \frac{\mathbf{W}_{p1} \odot \mathbf{X}_{p_1} + \mathbf{W}_{p2} \odot \mathbf{X}_{p_2}}{\mathbf{W}_{p1} + \mathbf{W}_{p2}} \tag{3}$$

where $\mathbf{W}_{p1}$ and $\mathbf{W}_{p2}$ are the weights maps yielded from the first and the second tunnel in the polymorphic transformer, respectively. $\mathbf{X}_{p_1}$ and $\mathbf{X}_{p_2}$ denote the outputs of these two tunnels, and $\odot$ represents the elementwise multiplication operation.

In the multistream morphological feature extraction, the size of feature maps remains consistent along with the overall processing procedure. On the other hand, the channel number changes across streams. For stream ①, the channel number remains unchanged. For stream ②, the descending-channel morphological module [Fig. 4(b)] halves the channel dimension, and the ascending-channel morphological module [Fig. 4(a)] doubles the channel dimension. While for stream ③, both the descending- and ascending-channel modules are applied twice, consecutively. After acquiring

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE BELGIUM DATA

Class		Number of Samples	
No.	Name	Training	Testing
1	Beech	80	321
2	Ash	13	54
3	Larch	23	93
4	Poplar	83	333
5	Copper beech	16	64
6	Chestnut	13	54
7	Oak	60	243
Total		288	1162

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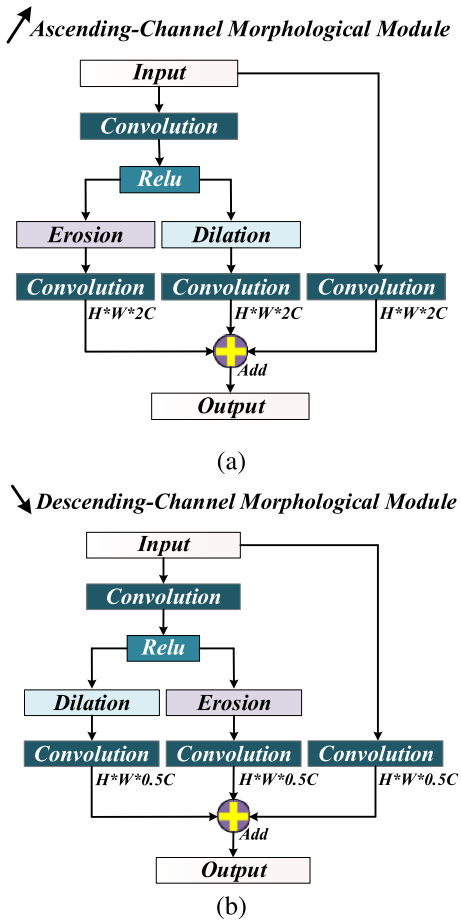


Fig. 4. Illustration of (a) ascending-channel morphological module and (b) descending-channel morphological module.

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE GAOFENG DATA

Class		Number of Samples	
No.	Name	Training	Testing
1	Cunninghamia lanceolata	4038	76705
2	Pinus massoniana	268	5072
3	Pinus elliottii	208	3967
4	Eucalyptus grandis x urophylla	746	14185
5	Eucalyptus urophylla	2014	38261
6	Castanopsis hystrix	1274	24187
7	Mytilaria laosensis	150	2850
8	Camellia oleifera	416	7909
9	Other broadleaf forest	538	10203
10	Road	524	9948
11	Cutting blank	508	9662
12	Building land	4	64
Total		10684	203012

diversified morphological features from the multistream propagation, a reliable feature aggregation method is needed here. The most commonly used strategy is feature concatenation, while it would restrict the expressive ability of the morphological extractor. Based on the self-attention mechanism, the dynamic fusion module is utilized to fuse features from different streams, as shown in Fig. 3. Particularly, the dynamic fusion process can help to realize the initial logical aggregation of features from the morphological space. Overall, the

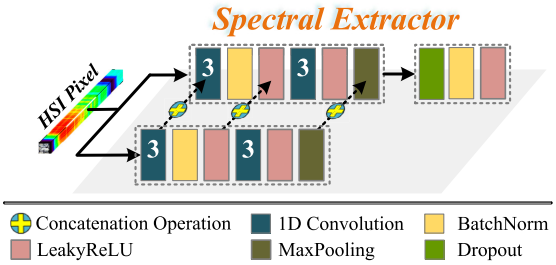


Fig. 5. Feature extraction process for F_{spe}.

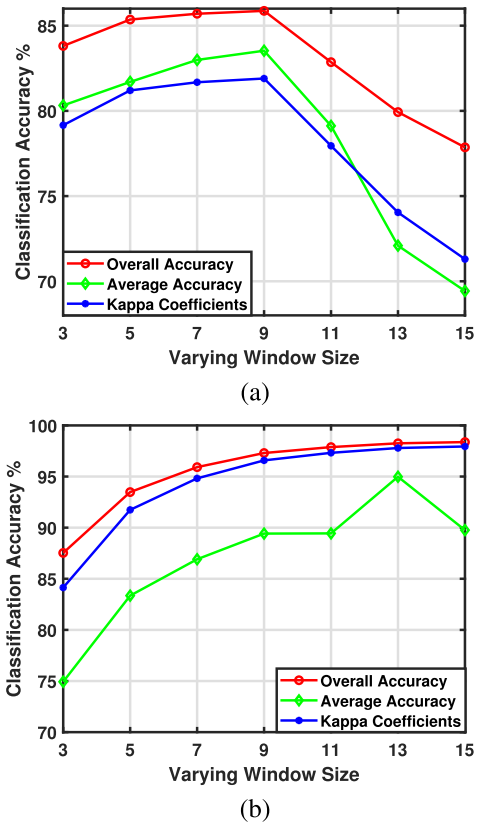


Fig. 6. Classification performance of the SLA-NET with varied input patch scales: (a) Belgium data and (b) Gaofeng data.

feature embeddings obtained by the morphological extractor are formulated by

$$\mathbf{F}_{\text{mor}} = \mathbf{E}_{\text{mor}}(\mathbf{X}). \tag{4}$$

B. Adaptive Fusion and Training Strategy

To integrate the spatial and morphological features, a trainable fusion module is developed. As shown in the third part of Fig. 2, $\mathbf{F}_{\text{spa}}$ and $\mathbf{F}_{\text{mor}}$ are first concatenated together

$$\mathbf{F}_{\text{conca}} = [\mathbf{F}_{\text{spa}}; \mathbf{F}_{\text{mor}}]. \tag{5}$$

Then, the combined features are utilized as a descriptor to recalibrate the feature maps derived from different extractors. The integrated features are denoted as

$$\mathbf{F}'_{\text{conca}} = [\mathbf{W}_{\text{spa}} \odot \mathbf{F}_{\text{spa}}; \mathbf{W}_{\text{mor}} \odot \mathbf{F}_{\text{mor}}] \tag{6}$$

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TABLE III
CLASSIFICATION PERFORMANCE OF TREE-SPECIES CLASSIFICATION WITH DIFFERENT MODULE SETTINGS ON BELGIUM DATA

Class		F_{spe}	F_{spe}+F_{spa}	F_{spe}+F_{mor}	SLA-NET (F_{spa}+F_{mor})
1	Beech	71.96	70.09	74.14	81.31
2	Ash	57.41	77.78	25.93	74.07
3	Larch	93.55	86.02	77.42	89.25
4	Poplar	81.08	97.90	94.89	94.89
5	Copper beech	100	100	96.88	100
6	Chestnut	55.56	64.81	51.85	62.96
7	Oak	40.08	82.64	54.96	82.23
OA (%)		69.77	83.72	74.33	**85.87**
AA (%)		71.38	82.75	68.01	**83.53**
Kappa (%)		62.39	79.25	66.65	**81.90**

TABLE IV
CLASSIFICATION PERFORMANCE OF TREE-SPECIES CLASSIFICATION WITH DIFFERENT MODULE SETTINGS ON GAOFENG DATA

Class		F_{spe}	F_{spe}+F_{spa}	F_{spe}+F_{mor}	SLA-NET (F_{spa}+F_{mor})
1	Cunninghamia lanceolata	83.34	99.06	97.00	98.91
2	Pinus massoniana	0.41	90.63	81.62	90.54
3	Pinus elliottii	0.3	97.78	76.68	98.64
4	Eucalyptus grandis x urophylla	76.72	100	88.76	99.77
5	Eucalyptus urophylla	63.32	99.91	98.26	100
6	Castanopsis hystrix	18.25	90.57	80.35	94.63
7	Mytilaria laosensis	5.82	100	61.93	99.79
8	Camellia oleifera	61.81	99.71	96.84	99.60
9	Other broadleaf forest	47.51	92.12	86.52	93.24
10	Road	96.88	98.91	97.88	99.05
11	Cutting blank	98.34	100	98.82	100
12	Building land	0	1.56	0	65.62
OA (%)		65.28	97.74	92.97	**98.25**
AA (%)		46.06	89.19	80.39	**94.98**
Kappa (%)		55.95	97.13	91.07	**97.79**

TABLE V
PERFORMANCE OF SLA-NET WITH OR WITHOUT DATA AUGMENTATION

Data	Metrics	Without Augmentation	With Augmentation
Beigium	OA (%)	80.96	85.87
	AA (%)	77	83.53
	Kappa (%)	75.53	81.90
Gaofeng	OA (%)	97.99	98.25
	AA (%)	91.05	94.98
	Kappa (%)	97.46	97.79

where $\odot$ represents the elementwise multiplication operation. $\mathbf{W}_{\text{spa}}$ and $\mathbf{W}_{\text{mor}}$ are the weights maps of spatial and morphological features, and they are derived by implementing the continuous operations of Conv-Softmax on $\mathbf{F}_{\text{conca}}$. Finally, fully connected layers and softmax layer are further applied to obtain the class label of the integrated features $\mathbf{F}'_{\text{conca}}$.

The training process of the SLA-NET is separated into two stages. In the first stage, the spatial extractor is trained from scratch, and the initial learning rate is set as 0.001. Particularly, Adam optimizer [46] is utilized, and the training process is implemented for 100 epochs. While in the second stage, all other parts depicted in Fig. 2 are adjusted together, where the pre-trained weights of the spatial extractor are reloaded, and other parts are still trained from scratch. The initial learning rate for the second stage is 0.0001. Besides, a simple data augmentation strategy is utilized during the training process, which generates additional samples without introducing extra labeling costs as in [47]. All weights and biases of the SLA-NET are initialized based on Glorot normalization [48], and the input data are all standardized. The standardization way of the input HSI is consistent with that in [49].

III. EXPERIMENTS AND ANALYSIS

In this section, two forest HSI datasets are utilized to evaluate the proposed SLA-NET. All the programs are conducted based on a personal computer equipped with Intel Core i9-10900K and with Nvidia 3080 GPUs. The main softwares are Tensorflow[1] and Keras.[2] Keras is the highly productive application programming interface of Tensorflow, and Tensorflow is an open-source machine learning platform that uses data flow graphs for capturing solutions.

A. Experimental Data

1) *Belgium Data* [49]: This data were acquired near the western part of Belgium, and this dataset recorded 1450 trees [53]. HSI comprises 286 spectral bands covering the range from 0.37 to 2.5 μm with a spatial resolution of 1.5 m. In particular, the tree-species in this dataset including Beech, Ash, Larch, Poplar, Copper beech, Chestnut, and Oak. Besides, the number of training and testing samples are listed in Table I.

[1]http://tensorflow.org/
[2]https://github.com/fchollet/keras

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TABLE VI

COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE BELGIUM DATA

No.	Performance												
	SVM [10]	ELM [11]	RF [12]	CNN [50]	RPCACNN [51]	SSCNN [26]	Context CNN [27]	CRNN [30]	DR-CNN [28]	BERT-HSI [52]	FusCNNFCN [34]	3DCNN [22]	SLA-NET
1	78.19	76.32	68.54	87.85	64.80	70.72	84.42	65.42	70.09	64.49	65.73	66.98	81.31
2	11.11	53.70	33.33	79.63	27.78	57.41	68.52	38.89	37.04	24.07	59.26	33.33	74.07
3	72.04	83.87	48.39	82.80	67.74	65.59	79.57	65.59	78.49	72.04	80.65	61.29	89.25
4	90.99	91.59	87.99	94.89	90.09	91.89	87.69	89.49	93.99	97.60	92.49	93.09	94.89
5	100	93.75	95.31	100	75	100	100	90.62	95.31	96.88	95.31	95.31	100
6	20.37	50	31.48	66.67	12.96	37.04	53.70	40.74	31.48	27.78	29.63	24.07	62.96
7	67.36	74.38	45.04	42.56	45.87	69.01	70.66	52.89	61.16	68.18	62.40	60.74	82.23
OA	74.50	79.59	65.72	79.33	64.77	75.45	80.79	68.73	73.82	73.56	73.56	70.71	**85.87**
AA	62.87	74.80	58.58	79.20	54.89	70.24	77.79	63.38	66.80	64.43	69.35	62.12	**83.53**
Kappa	66.72	73.72	55.26	73.33	54.07	68.33	75.29	59.92	66.12	65.67	65.89	62.18	**81.90**

TABLE VII

COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE GAOFENG DATA

No.	Performance												
	SVM [10]	ELM [11]	RF [12]	CNN [50]	RPCACNN [51]	SSCNN [26]	Context CNN [27]	CRNN [30]	DR-CNN [28]	BERT-HSI [52]	FusCNNFCN [34]	3DCNN [22]	SLA-NET
1	90.03	89.86	90.69	86.63	83.48	91.61	99.58	96.78	98.00	96.99	95.78	95.43	98.91
2	10.88	2.94	14.33	11.73	34.27	75.51	97.22	90.40	92.61	92.94	89.33	92.84	90.54
3	5.27	0.53	7.94	3.05	20.04	85.53	100	99.60	92.74	93.80	84.93	97.08	98.64
4	52.39	50	57.12	77.99	52.23	86.07	98.10	99.99	99.28	97.46	98.06	96.84	99.77
5	69.99	69.53	72.76	61.86	66.16	96.73	99.98	98.92	99.58	92.44	98.48	97.51	100
6	40.00	40.29	37.01	45.59	29.80	85.07	82.43	87.95	86.41	90.09	83.82	84.41	94.63
7	10.32	7.79	16.60	1.16	6.91	70.28	100	98.21	96.91	83.58	92.67	94.28	99.79
8	55.22	46.93	64.53	58.80	72.03	95.52	99.65	99.43	99.06	99.97	97.77	98.47	99.60
9	35.56	41.60	43.78	45.53	54.82	68.81	92.29	96.20	89.16	93.46	83.40	90.74	93.24
10	97.75	97.02	97.56	97.63	98.13	99.10	99.59	98.95	99.76	99.69	99.49	99.01	99.05
11	90.92	95.82	95.56	94.78	95.74	99.15	100	99.98	99.83	98.34	99.81	99.69	100
12	0	6.25	0	7.81	0	90.63	71.88	95.31	26.56	68.75	48.44	45.31	65.62
OA	69.21	68.78	71.17	69.59	67.48	90.32	97.11	96.60	96.50	95.12	94.42	94.80	**98.25**
AA	46.53	45.71	49.82	49.40	51.13	87	95.06	**96.81**	89.99	92.29	89.33	90.97	94.98
Kappa	59.50	58.84	62.28	60.91	58.69	87.79	96.33	95.72	95.57	93.84	92.95	93.45	**97.79**

TABLE VIII

ELAPSED TIME (S: SECOND) OF TRAINING AND TESTING PROCESS FOR DIFFERENT METHODS ON BELGIUM DATA

	Training time (s)	Testing time (s)
CNN	29.27	50.12
SSCNN	24.24	62.78
Context CNN	44.26	77.02
DR-CNN	90.14	131.93
BERT-HSI	135.44	168.84
FusCNNGCN	19.12	71.43
3DCNN	113.61	102.27
SLA-NET	89.14	185.40

2) *Gaofeng Data* [22]: The second dataset was acquired by the AISA Eagle II diffraction grating push-broom hyperspectral imager over Jiepai branch of Gaofeng State Owned Forest Farm in Guangxi province in China. In this data, there are 125 spectral bands covering the range from 0.4 to 1 μm with a spectral resolution of 3.3 nm and a spatial resolution of 1 m. The forest in this study area is mainly man-made forest. Table II lists the number of training and testing samples.

B. Parameter Tuning and Module Analysis

The fine-grained classification performance of tree species is closely correlated with the architecture of the designed framework. To evaluate the effectiveness of the proposed SLA-NET, the classification performance with using different module settings is analyzed in this section.

1) Impact of Diversified Features Extraction: The joint use of spectral and spatial features has been validated to be effective for boosting HSI classification performance in a great deal of literature. While in the classification task of forestry tree species, the spectral responses of different plants are extremely similar, and the coarse spatial textures of the same class vary considerably. Accordingly, in the proposed framework, micro-morphological structures are meticulously retained and integrated by the spatial-logical aggregation design. Experiments are implemented to assess the impact of diversified information utilization. Specifically, the performance of "F_{spe}," "$F_{\text{spe}} + F_{\text{spa}}$," "$F_{\text{spe}} + F_{\text{mor}}$," and "SLA-NET ($F_{\text{spa}} + F_{\text{mor}}$)" is studied. The spectral feature extraction process for F_{spe} is depicted in Fig. 5. As listed in Tables III and IV, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are used to evaluate the classification performance. When the tree-species classification process is implemented without morphological and spatial features, the incomplete spatial information flow decreases the ability of diagnostic identification of fine-grained tree species. In the Belgium dataset, the classification performance of "F_{spe}" is clearly less effective than that of other baselines. Meanwhile, with both

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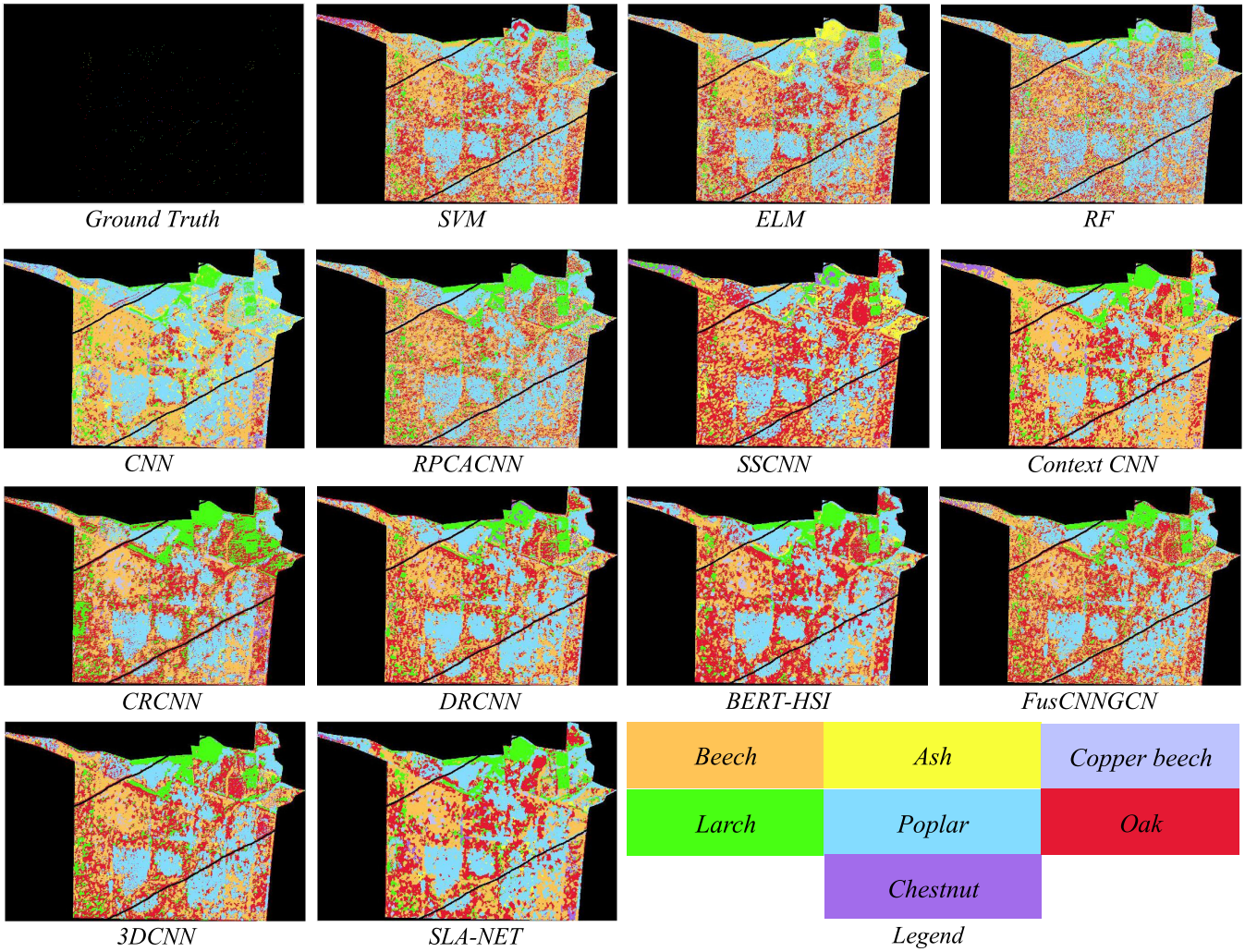


Fig. 7. Classification maps for the Belgium data with different methods including: SVM (74.50%), ELM (79.59%), RF (65.72%), CNN (79.33%), RPCACNN (64.77%), SSCNN (75.45%), Context CNN (80.79%), CRNN (68.73%), DRCNN (73.82%), BERT-HSI (73.56%), FusCNNGCN (73.56%), 3-DCNN (70.71%), and SLA-NET (85.87%).

spectral and incomplete spatial information, the performance of "$F_{\text{spe}} + F_{\text{spa}}$" and "$F_{\text{spe}} + F_{\text{mor}}$" is still less competitive than that of SLA-NET. Similarly, the spatial extraction process that removes either the coarse-grained or the fine-grained extractor reduces the ability to complete spatial information acquisition and results in lower classification performance. Based on the comparisons provided in Tables III and IV, the SLA-NET provides the best classification performance in the fine-grained classification task.

2) Multiscale Comparison: The impact of input patch size on the classification performance is then evaluated. The performance on different input sizes, i.e., from 3×3 to 15×15, is illustrated in Fig. 6. In particular, related results clearly suggest that the input patches with varied sizes result in strikingly different performance. To be specific, for the Belgium dataset, the patch size of 9×9 provides the best classification performance. For the Gaofeng dataset, by contrast, the OA is increased with the increase of input size, and the input sizes of 13×13 and 15×15 achieve similar classification performance. Here, we choose a relatively small size (i.e., 13×13) as the input size for the Gaofeng dataset to save computational resources. Therefore, the input patch sizes of the two datasets are set as 9 and 13, respectively, for the subsequent experiments. Based on the aforementioned analysis, it can be seen that the selection of the input size relies on a manual process, and the input size has a great influence on the classification result. There is still potential for improvement in automating input selection.

3) Impact of Data Augmentation: A deep network usually requires a significant amount of training data to learn the model with many parameters. However, in practice, only limited samples are available for the classification task. In this article, we employ a simple data augmentation strategy as mentioned in Section II-B. Table V evaluates the classification performance acquired with or without the data augmentation process. In the Belgium dataset, the classification process with data augmentation yields approximately 4.91% improvements over that without augmentation setting. In the Gaofeng data, the classification process with the augmentation setting shows approximately 0.26% improvements over that without

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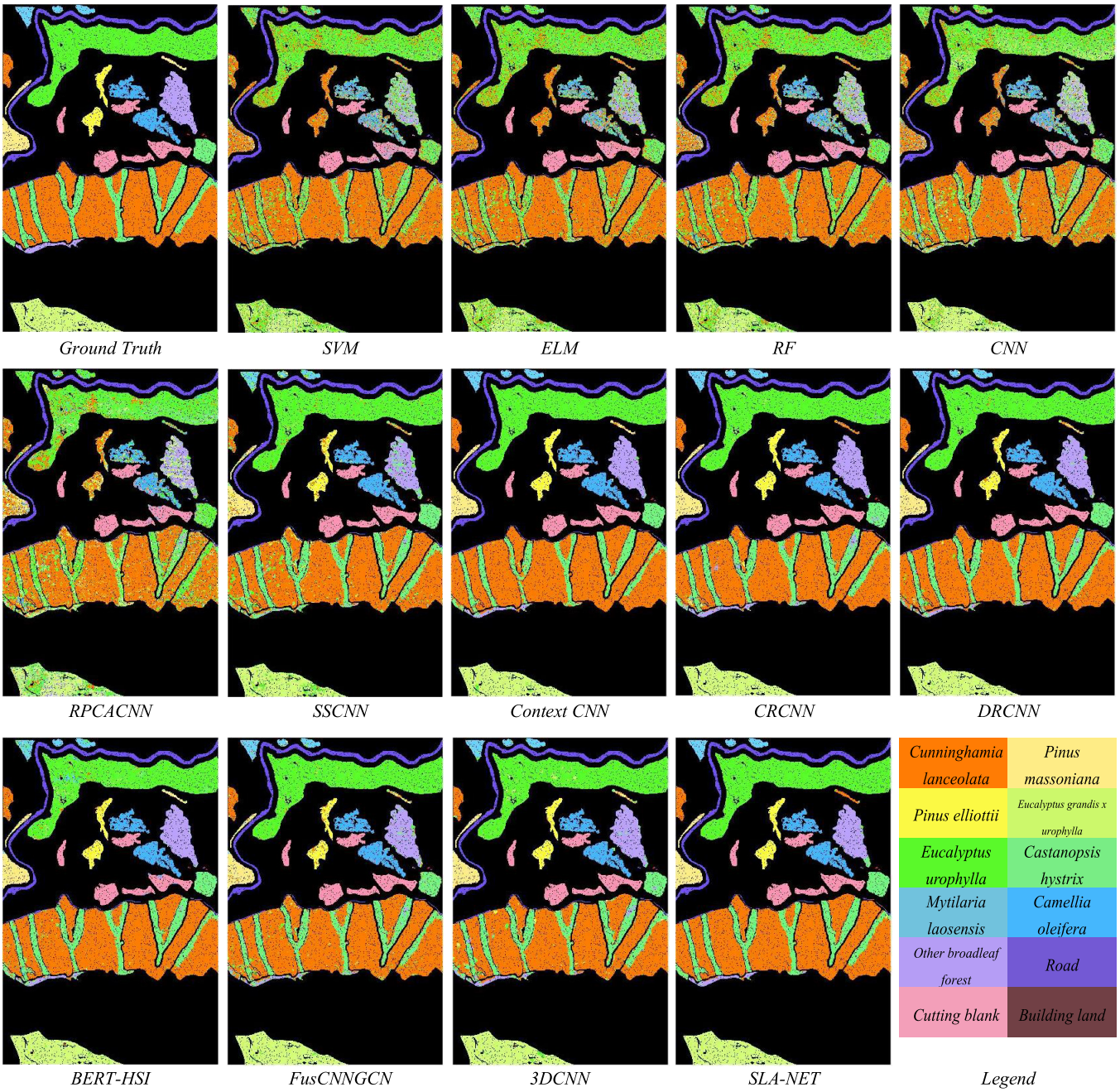


Fig. 8. Classification maps for the Gaofeng data with different methods including: SVM (69.21%), ELM (68.78%), RF (71.17%), CNN (69.59%), RPCACNN (67.48%), SSCNN (90.32%), Context CNN (97.11%), CRNN (96.60%), DRCNN (96.50%), BERT-HSI (95.12%), FusCNNGCN (94.42%), 3-DCNN (94.80%), and SLA-NET (98.25%).

augmentation operation. It is clear that the training data augmentation process is positive for improving the fine-grained classification performance.

C. Classification Performance and Analysis

To evaluate the effectiveness of the proposed SLA-NET, we compare it with both traditional methods (e.g., SVM [10], ELM [11], and RF [12]) and some deep learning-based methods, including CNN [50], RPCACNN [51], SSCNN [26], context CNN [27], CRNN [30], DRCNN [28], BERT-HSI [52], FusCNNGCN [34], and 3-DCNN [22]. In particular, all the comparison methods are implemented with the recommended settings of the corresponding literature.

In Tables VI and VII, for all experiments on the aforementioned two datasets, the performance of RPCACNN is generally worse than those of full-spectrum-related methods. Here, RPCACNN yields the worst classification performance, and the primary reason is that randomized PCA is employed on HSI input beforehand, which destroys the diagnosis information within HSI data. Besides, it is clear that the proposed SLA-NET is superior to other methods. Using the *Belgium*

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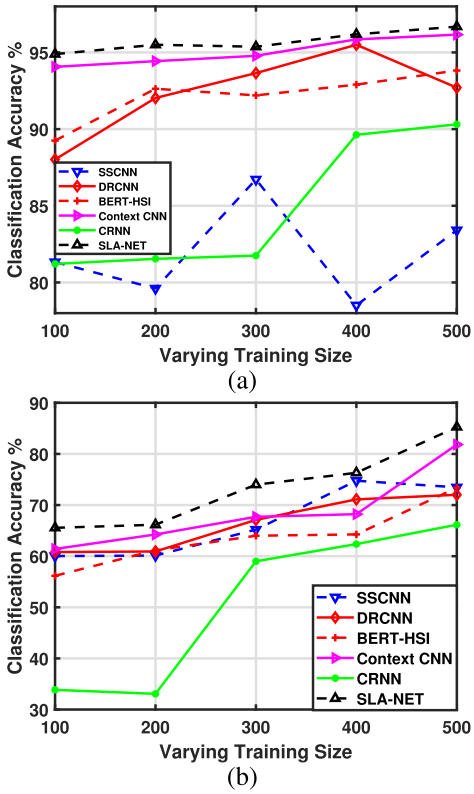


Fig. 9. Classification performance with different sizes of training samples using: (a) Belgium data, (b) Gaofeng data.

dataset for example, it shows approximately 11.37%, 6.28%, 20.15%, 6.54%, 21.1%, 10.42%, 5.08%, 17.14%, 12.05%, 12.31%, 12.31%, and 15.16% improvements over SVM, ELM, RF, CNN, RPCACNN, SSCNN, Context CNN, CRNN, DR-CNN, BERT-HSI, FusCNNGCN, and 3-DCNN, respectively. For the *Gaofeng dataset*, it yields approximately 29.04%, 29.47%, 27.08%, 28.66%, 30.77%, 7.93%, 1.14%, 1.65%, 1.75%, 3.13%, 3.83%, and 3.45% improvements over SVM, ELM, RF, CNN, RPCACNN, SSCNN, Context CNN, CRNN, DR-CNN, BERT-HSI, FusCNNGCN, and 3-DCNN, respectively. Accordingly, it is clear that the proposed spatial-logical aggregation design achieves better classification performance than other methods due to more effective spatial logic integration. In particular, the classification results of SLA-NET on the *Chestnut* and *Oak* classes are significantly higher than other baselines. This is because the SLA-NET is directed toward the distribution characteristics of most tree species. The Chestnut- and Oak-species groups usually take on a broad round-like look, and they inevitably show uneven textures from the satellite view. Accordingly, it is important to design feature extractors with both coarse- and fine-grained perception scales for these tree species recognition. In this regard, the proposed SLA-NET assembles spectral, spatial, and morphological details based on the multitunnel aggregation network and the dynamic fusion modules, which can better improve the spatial perception ability of the feature extraction process, thus assisting in a more accurate classification of these tree species, i.e., Chestnut trees and Oak trees.

Based on the comparisons listed in Tables VI and VII, the effectiveness of SLA-NET is confirmed. To facilitate comparison between different baselines, Figs. 7 and 8 further illustrate the classification maps. The classification results of RPCACNN contain a large number of salt-and-pepper pixels, which is also consistent with the poor classification performance listed in Tables VI and VII. In contrast, the proposed method produces visually better classification results with fewer mislabeled pixels than other baselines. Even in comparison to the classification maps from the most competitive baseline, SLA-NET can also obtain more accurate results. Taking the *Belgium data* for an example, the proposed method obtains a spatially smoother map, e.g., the *Larch* class in Fig. 7. Moreover, the classification performance of SSCNN, Context CNN, CRNN, DR-CNN, and BERT-HSI is also competitive in all datasets.

To further demonstrate the effectiveness of SLA-NET, Fig. 9 shows the sensitivity of the competitive methods versus different training samples size. For the Belgium dataset, the number of samples in different categories varies in the set of $\{5, 10, 20, 40, 80\}$. In the Gaofeng dataset, the number of samples in different categories varies in the set of $\{100, 200, 300, 400, 500\}$. It should be noted that all the training samples are chosen from a random selection. For an individual class, if the number of training samples does not meet the quantity demand, all samples in this class are utilized in the training set. Obviously, compared with other comparison methods, SLA-NET yields consistently better results versus varied sample sizes. For example, in Fig. 9(b), when the training set is set as 100 samples per class, the overall accuracy for the SLA-NET is approximately 94.89%, while the classification accuracy for most of the other methods is all below 90%. The performance improvement is remarkable, which also verifies the effectiveness and robustness of the SLA-NET in diversified sample settings. Furthermore, Table VIII summarizes the training and testing computational cost of some competitive methods on Belgium data. For this part of the experiment, all the algorithms were carried out using Pytorch on an AMD Threadripper 3955wx 16-Core Processor (128-GB RAM) powered with Nvidia GTX 3080 GPU. Among all baselines, BERT-HSI takes the longest time for model training and testing process. The computational cost of SLA-NET is lower than that of DR-CNN, BERT-HSI, and 2-DCNN. Also, the proposed SLA-NET consists of several steps of self-defined operations, and there is still potential for further reducing the complexity. Based on the classification accuracy and the computational cost, SLA-NET yields more satisfying performance over other baselines.

IV. Conclusion

In this article, a fine-grained tree species classification method, called SLA-NET, was proposed to integrate spectral, spatial, and morphological details in HSI based on several plug-and-play dynamic-fusion modules. The SLA-NET-integrated spatial convolutional and morphological operations into a unified architecture, which provided an effective solution to coarse-to-fine feature extraction, reducing the information loss of data transmission in the deep network. In particular, to

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further preserve the desired fine spatial details, the multistream morphological feature extraction module is deployed with both dual-mode transmission and self-attention guided feature fusion. The experiments on two forestry datasets confirmed that the proposed SLA-NET can provide superior performance when compared with other state-of-the-art methods.

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Mengmeng Zhang received the B.S. degree in computer science and technology from the Qingdao University of Science and Technology, Qingdao, China, in 2014, and the Ph.D. degree in control science and engineering from the Beijing University of Chemical Technology, Beijing, China, in 2019.

She is currently a Post-Doctoral Researcher with the School of Information and Electronics, Beijing Institute of Technology, Beijing. Her research interests include remote sensing image processing and pattern recognition.



Wei Li (Senior Member, IEEE) received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2007, the M.S. degree in information science and technology from Sun Yat-Sen University, Guangzhou, China, in 2009, and the Ph.D. degree in electrical and computer engineering from Mississippi State University, Starkville, MS, USA, in 2012.

He spent 1 year as a Post-Doctoral Researcher with the University of California at Davis, Davis, CA, USA. He is currently a Professor with the School of Information and Electronics, Beijing Institute of Technology, Beijing, China. His research interests include hyperspectral image analysis, pattern recognition, and data compression.

Dr. Li is an Associate Editor of the IEEE SIGNAL PROCESSING LETTERS (SPL) and the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS).



Xudong Zhao (Student Member, IEEE) received the B.S. degree in science and technology from the Electronic Information Department, Beijing Institute of Technology, Beijing, China, in 2016, where he is currently pursuing the Ph.D. degree in information and communication engineering and the second Ph.D. degree in computer science engineering with Ghent University, Ghent, Belgium.

His research interests include fractional signal processing and multisource remote sensing.



Huan Liu received the B.S. degree from the China University of Mining and Technology, Xuzhou, China, in 2013, and the M.S. and Ph.D. degrees from the Beijing Institute of Technology, Beijing, China, in 2016 and 2022, respectively.

His research interests include image processing, image classification, and transfer learning.



Ran Tao (Senior Member, IEEE) was born in 1964. He received the B.S. degree from the Electronic Engineering Institute of PLA, Hefei, China, in 1985, and the M.S. and Ph.D. degrees from the Harbin Institute of Technology, Harbin, China, in 1990 and 1993, respectively.

In 2001, he was a Senior Visiting Scholar with the University of Michigan, Ann Arbor, MI, USA. He is currently a Professor with the School of Information and Electronics, Beijing Institute of Technology, Beijing, China. He has three books and over 100 peer-reviewed journal articles. His research interests include fractional Fourier transform and its applications, theory, and technology for radar and communication systems.

Dr. Tao is a member of the Wireless Communication and Signal Processing Commission of International Union of Radio Science (URSI). He was a recipient of the National Science Foundation of China for Distinguished Young Scholars in 2006, and the First Prize of Science and Technology Progress in 2006 and 2007, and the First Prize of Natural Science in 2013, both awarded by the Ministry of Education. He was a Distinguished Professor of the Changjiang Scholars Program in 2009. He was a Chief Professor of the Program for Changjiang Scholars and Innovative Research Team in University from 2010 to 2012. He has been a Chief Professor of the Creative Research Groups of the National Natural Science Foundation of China since 2014. He is the Vice Chair of the IEEE China Council and the URSI China Council. He is an Associate Editor of the IEEE SIGNAL PROCESSING LETTERS (SPL).



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore, MD, USA, in 2000.

She is currently the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of the SPIE—International Society for Optics and Photonics. She received the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. She was a Co-Chair of the Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society from 2009 to 2013 and the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She is a General Chair of the 4th IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing, Shanghai, China, in 2012. She has served as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the JOURNAL OF APPLIED REMOTE SENSING, and the IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

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